

# Anmol Bhandohal

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**BSc Electrical Engineering, Co-op, University of Alberta** • Class of 2029 • 5 of 8 academic terms complete

**Available for co-op: January – August 2027 (8 months, or two 4-month terms)**

## TECHNICAL SKILLS

**Design & EDA:** Altium Designer, LTspice, Fusion 360 (CAD), PCB Design (2 & 4 layer), STM32CubeIDE

**Programming:** C/C++, Python, MATLAB, VHDL, Git

**Protocols:** I2C, SPI, UART, PWM, CAN bus

**Lab & Hardware:** Oscilloscope, Multimeter, Soldering (through-hole & SMD, rework), 3D Printing, Zynq Z7 FPGA

**Field & Code:** Canadian Electrical Code (CEC), Fire alarm programming & verification (Mircom), 347 V / 120/208 V systems, Conduit & panel termination, Blueprint reading

## WORK EXPERIENCE

**Electrical Apprentice — Commercial & Residential Construction** — *Powerworks, Edmonton, AB* **Summer 2026**

- Programmed a Mircom FX-3500 addressable fire alarm system for a \$9.6M three-storey commercial building, configuring pull stations, smoke and heat detectors, and elevator recall relay logic so cars automatically return to a safe floor and hold on alarm
- Isolated an intermittent ground fault on a live fire alarm circuit by half-split isolation across the panel's pluggable terminal blocks, ohming out each branch for continuity until the faulted segment was localized — restoring service without pulling new cable
- Interpreted multi-page commercial drawings — single-line diagrams, panel schedules, and device layouts across three levels — and directed crew on install sequence and circuit routing from those prints
- Walked a completed commercial building with the consulting engineer for verification and acceptance inspection, confirming device operation and clearing deficiencies before sign-off
- Completed seven sites in one term across three commercial and four residential builds — retail plaza, daycare, 15-unit townhouse duplexes, 6-plexes — plus a hockey arena power upgrade and temporary generator distribution keeping a commercial site energized through construction
- Installed and terminated 347 V and 120/208 V services, disconnects, and multi-meter distribution stacks; bent and ran conduit, pulled feeders with fish tape and vacuum, strapped BX and TECK, and installed lighting, HRV/heater fans, receptacles, and low-voltage data and telecom to Canadian Electrical Code

## ENGINEERING TEAMS & COMPETITIONS

**Electrical Subsystem Engineer** — *ARVP — Autonomous Robotic Vehicle Project, University of Alberta* **Sept. 2025 – Present**

- Revised and documented the Comms Hub PCB in Altium, replacing pin headers with keyed shrouded connectors for reverse-polarity protection and activating CAN bus transceivers for vehicle communications
- Designed a 4-layer PCB stackup in Altium with dedicated signal, ground, and power planes, applying manufacturer clearance constraints and clearing a full DRC with zero violations before sign-off
- Designed an active op-amp low-pass filter in LTspice to isolate a 1 kHz signal from 15 kHz noise, validating frequency response through simulation before implementation
- Supported in-water testing of an autonomous underwater vehicle, diagnosing and resolving hardware faults on-site in real time

**GeoWall Engineering Design Competition — First Place** **Feb. 2025**

- Placed first against 4th-year Civil Engineering teams, leading structural planning two weeks ahead through drawings, material selection, and kPa load predictions; the design sustained the competition's highest recorded hydraulic press load
- Earned Best Presentation across all competing teams

## ENGINEERING PROJECTS

**ESP32 Environmental Monitor** | *C++, ESP32, I2C, REST API, JSON* **Fall 2025**

- Built an ESP32 environmental monitor in C++ sampling temperature, humidity, pressure, and ambient light over I2C into a 720-point rolling buffer sized for a continuous 6-hour capture
- Developed a Wi-Fi HTTP server exposing a live auto-refreshing dashboard, REST endpoints serving real-time JSON, and CSV export for offline analysis
- Implemented a labelled session workflow via a /setroom endpoint to tag and reset capture runs for clean comparison between test locations

**Adaptive Closed-Loop LED Lamp** | *C++, ESP32, PWM, MOSFET driver, Altium* **In progress**

- Designing a closed-loop illuminance controller on ESP32 that holds a target lux level by PWM-driving a 12 V LED strip against changing ambient light, using a low-side MOSFET driver with gate resistor and pulldown for clean 5 kHz switching
- Laying out a custom 2-layer carrier PCB in Altium with a full ground plane and dedicated subsystem headers to replace the breadboard prototype

## CERTIFICATIONS & ADDITIONAL

EECERT Lab Certification — U of A (Circuit Design, PCB Layout, Arduino) • First Class Standing • *Class 5 Licence* | *Willing to relocate* | *Field-ready*